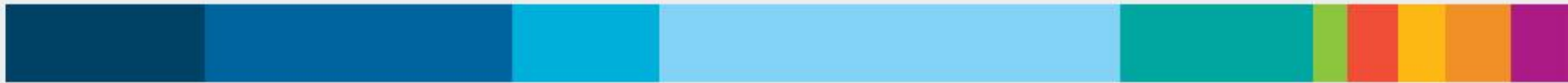




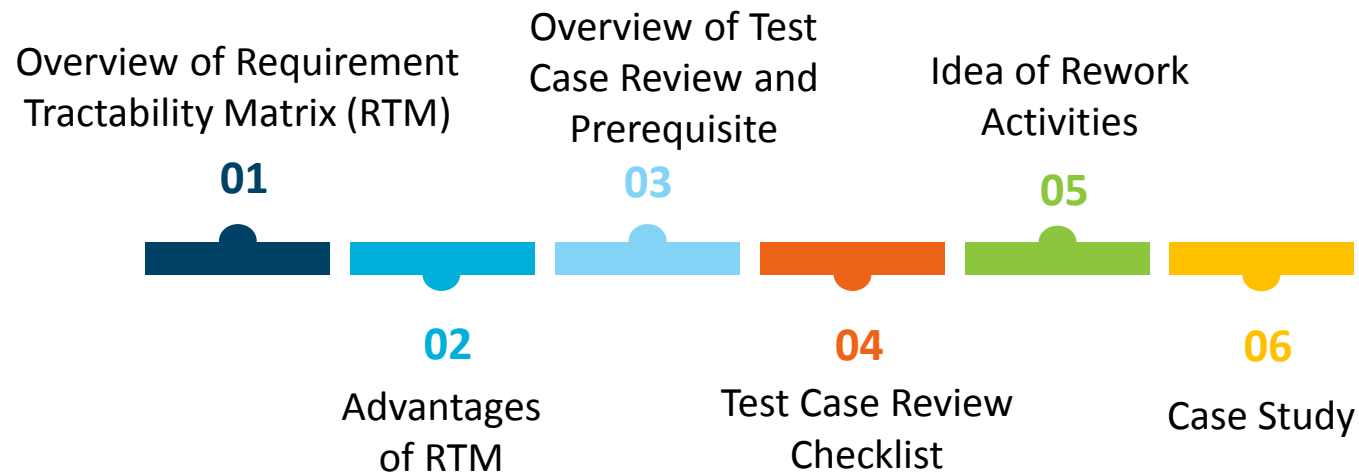
Basic Testing



Day 6: Review, and Rework of Test Cases



Day 06



01 Overview of Requirement Traceability Matrix (RTM)

What is a Traceability Matrix (RTM)?



Traceability Matrix

- A document that co-relates any two-baseline documents requiring a many-to-many relationship to **check the completeness of the relationship.**
- It is used to **track the requirements** and to **check whether the current project requirements are met.**

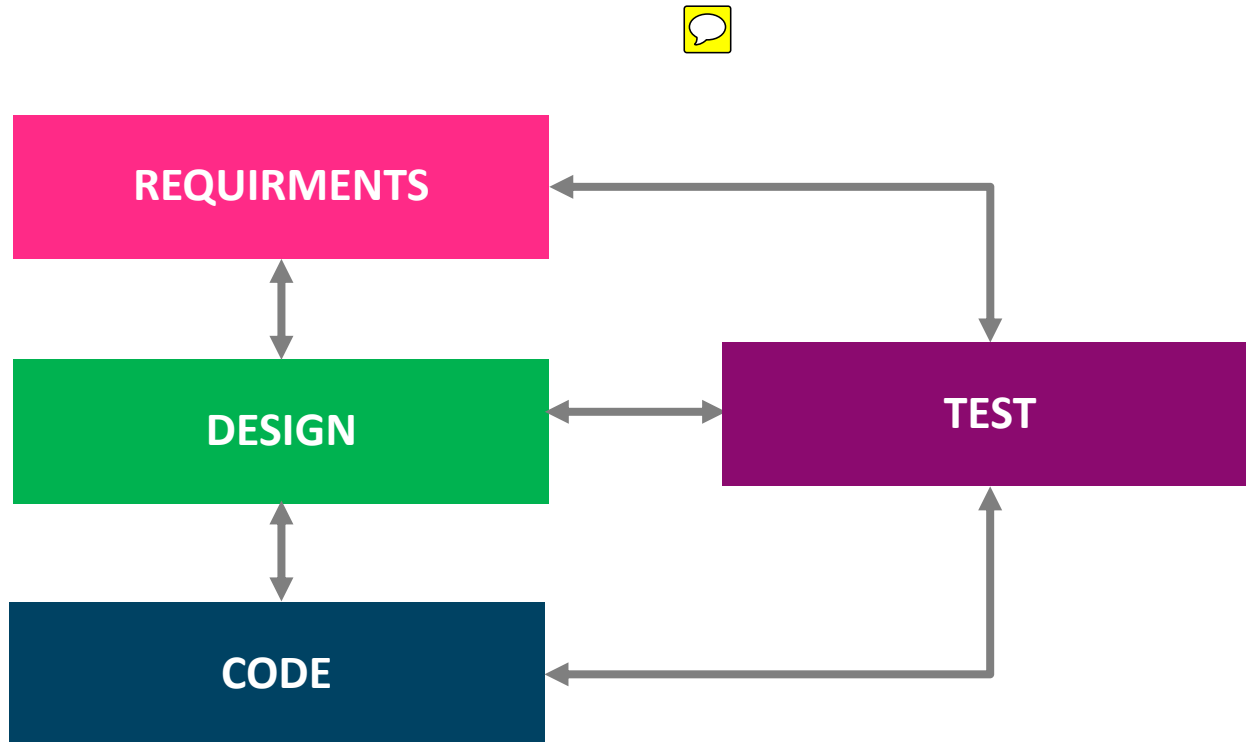
Overview of Requirement Traceability Matrix (RTM)



Requirement Traceability Matrix (RTM)

- **Captures all requirements** proposed by the client or development team and their traceability in a single document delivered at the conclusion of the lifecycle.
- It is a document **mapping and tracing users requirement** with test cases.
- The main purpose of RTM is to **see that all test cases are covered** so that no functionality is missed during testing.

Structure of a Simple RTM





Questions?

01

What is a Requirements Traceability Matrix (RTM)?

A

A document that co-relates any two-baseline documents requiring a many-to-many relationship to check the completeness of the relationship.

B

It is a document mapping and tracing users requirement with test cases.

C

It is a document describing the testing scope and activities. It is the basis for formally testing any software/product in a project.

D

It is a document that contains summary of test activities and test results.

02 Advantages of RTM

What are the advantages of RTM?



- 1 It provides a **single view of all the project documents** that map to specific project requirements.
- 2 It provides the ability to **describe and follow the testing of a requirement both initially and after changes** that impact them.
- 3 **All documents** developed during the design, code and testing processes in subsequent lifecycle stages must be **traced back to the project requirements (consistency)**.
- 4 This traceability assures that **the product will satisfy all of the requirements and remain within the project scope**, including testing boundary conditions and alternative scenarios (**completeness**).
- 5 It provides a **single source to clearly identify and manage scope creep**.
- 6 It can **identify appropriate test cases and scenarios** for optimal coverage.

What are the advantages of RTM? (continued)



- 7 It confirms **100% test coverage**.
- 8 It highlights any requirements **missing or document inconsistencies**.
- 9 It shows the **overall defects or execution status** with a focus on business requirements.
- 10 It helps in **analyzing or estimating the impact on the QA team's work** with respect to revisiting or re-working on the test case.
- 11 It makes it **easy to identify the missing functionalities**.
- 12 If there is a change request for a requirement, then we can **easily find out which test cases need to be updated**.

A c t i v i t y

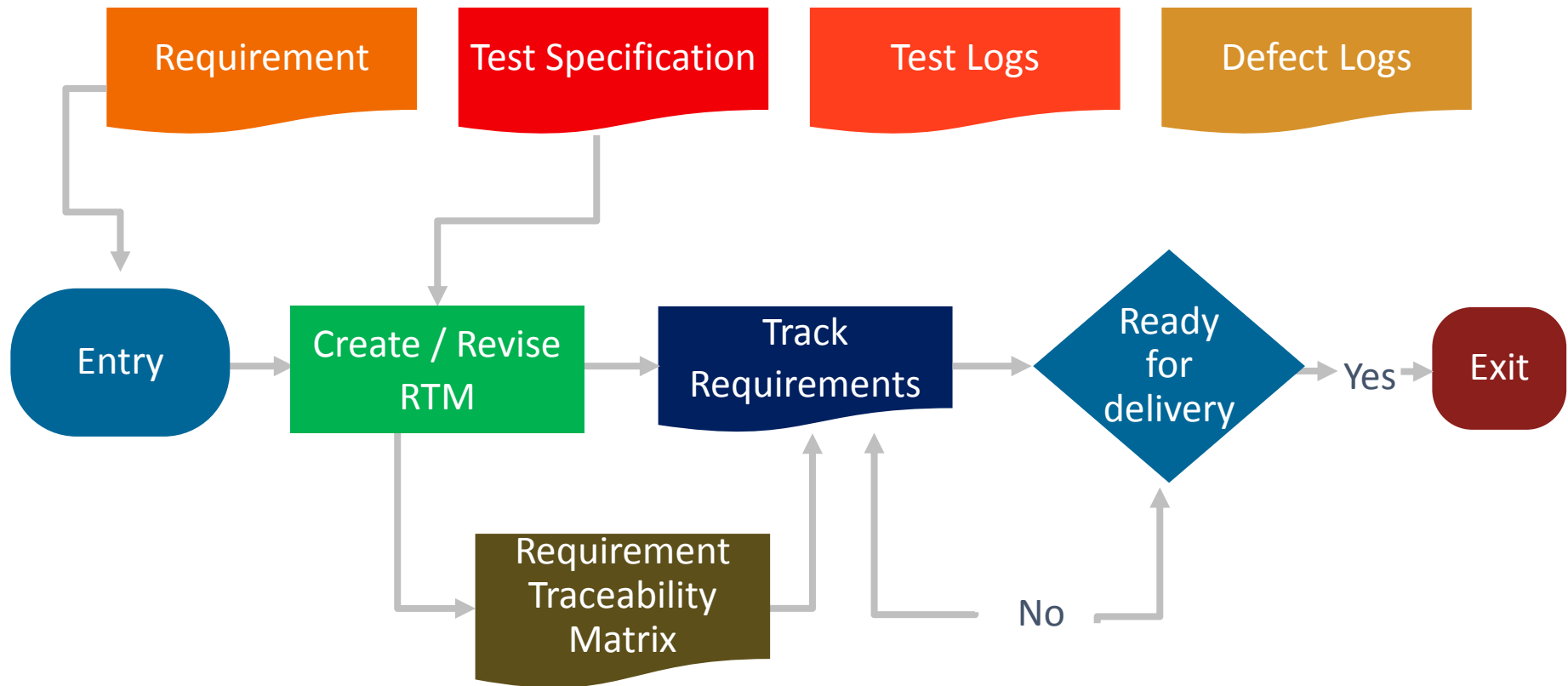
Use a real case and create Requirement Traceability Matrix.



Requirement Tracability Matrix



Mapping Process of RTM





Goals of RTM

Ratify the system architecture.

Establish a basis for subsequent component architecture.

Provide traceability of requirement through the lifecycle.

Use Cases	Functionality	Test Cases
UC 1.1.1	Create a User input box.	TC.1.1.1
UC 1.1.2	Create a Password input box.	TC.1.1.2
UC 1.1.3	Create a Submit button.	TC.1.1.1 TC 1.1.2.TC 1.1.3
UC 1.1.4	Create a Cancel button	TC.1.1.1.4
UC 1.1.5	Submit it button takes only desired valid inputs.	TC.1.1.1.5

02

Which of the following statement is true as an advantage of RTM?

A

It confirms most of the test coverage.

B

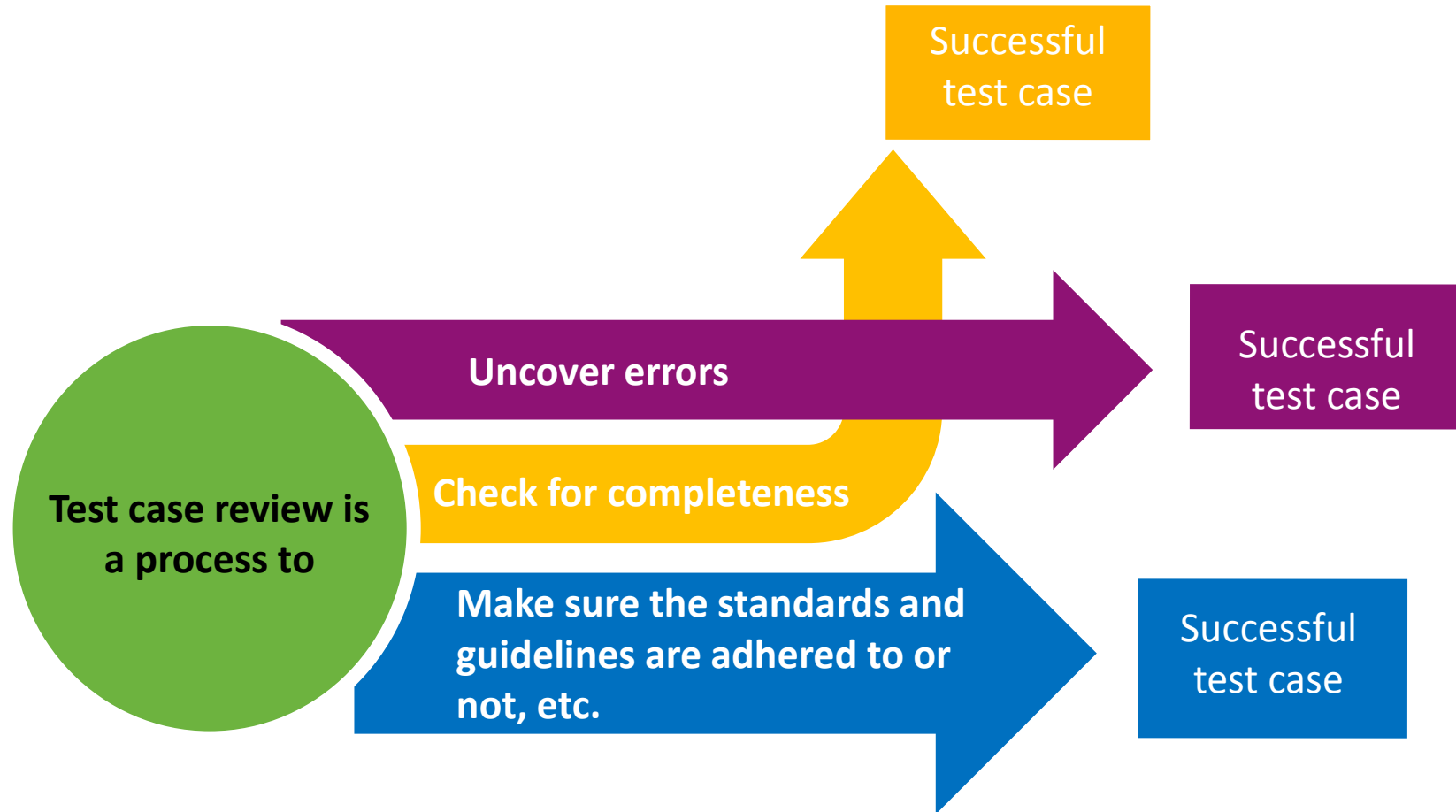
All documents developed only during the design process can be traced back to the requirements.

C

It helps in analyzing or estimating the impact on the QA team's work concerning revising the test case.

03 Overview of Test Case Review and Prerequisite

Why do we use Test Case Reviews?



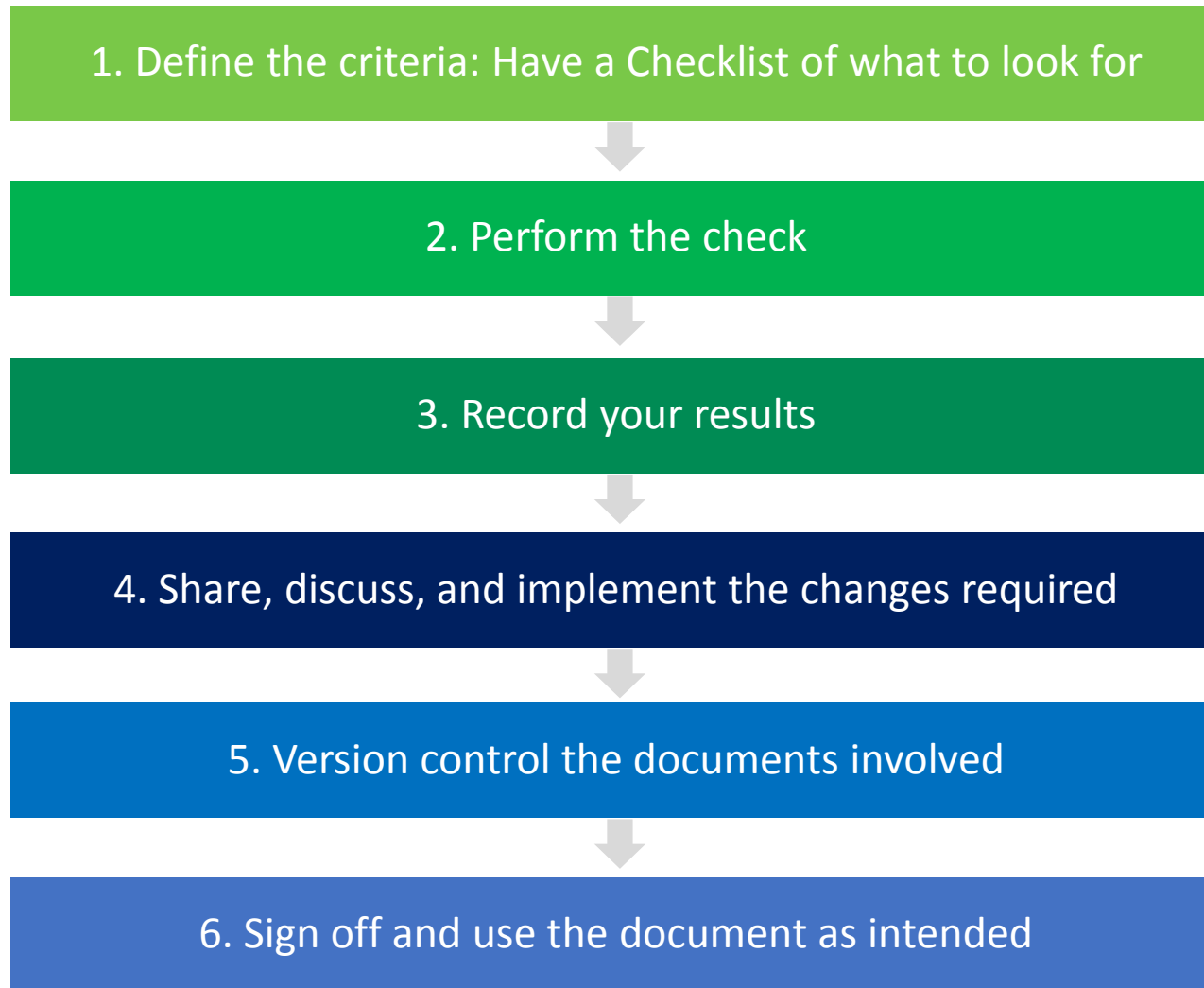
Benefits of Test Case Reviews?



**Defect prevention
while reviewing
requirements**

**Conceptual and
Technical Coverage**

**Defect
prevention while
reviewing test
cases**





Questions?

03

Which of the following statement is true as a benefit of conducting a test case review?

A

It helps prevent defects while reviewing requirements.

B

It assures conceptual and technical coverage.

C

It helps prevent defects while reviewing test cases.

C

It helps prevent defects while reviewing test cases.

04 Test Case Review Checklist

Test Case Review Checklist: A Sample (1 of 3)



Here is a sample showing the key checks that are covered in Test case review Checklists.

- ✓ All the requirements mentioned in Functional Requirement Specification (FRS) .
- ✓ All negative scenario tests are also covered.
- ✓ Boundary value conditions are covered. That is, tests covering lower/upper bounds are covered.
- ✓ Data validity tests are covered.
- ✓ All the GUI related test cases (if mentioned in FRS) are covered.



Test Case Review Checklist: A Sample (2 of 3)



- ✓ To check if there is any invalid test cases
- ✓ To check if there is any redundancy in test cases
- ✓ Objective of TC, design, preconditions, post conditions, expected result, actual result
- ✓ To check that the test case priority
- ✓ To check narration of test case





- ✓ The purpose of a checklist is to check whether no major scenarios are missing in test cases
- ✓ Test steps are precise and easily understandable
- ✓ Clear expected result is mentioned for each step
- ✓ Check for all text or grammatical errors
- ✓ The number of test steps should not be more than 10
- ✓ The Test Case Review has Information related to setup of test environment and pre-requisites



- WPI(Work product Inspection) should be sent to test case author for rework.
- Author can accept or reject review comments with proper justification.
- Once author accept, he/she should modify test cases as per review comments.
- Author should update issue log accordingly.
- Author should send issue log and updated test cases for another review.
- Once rework is done and no further review comments are given by reviewer, test cases status should be ready for execution.
- WPI should kept in repository for future reference / quality audit.

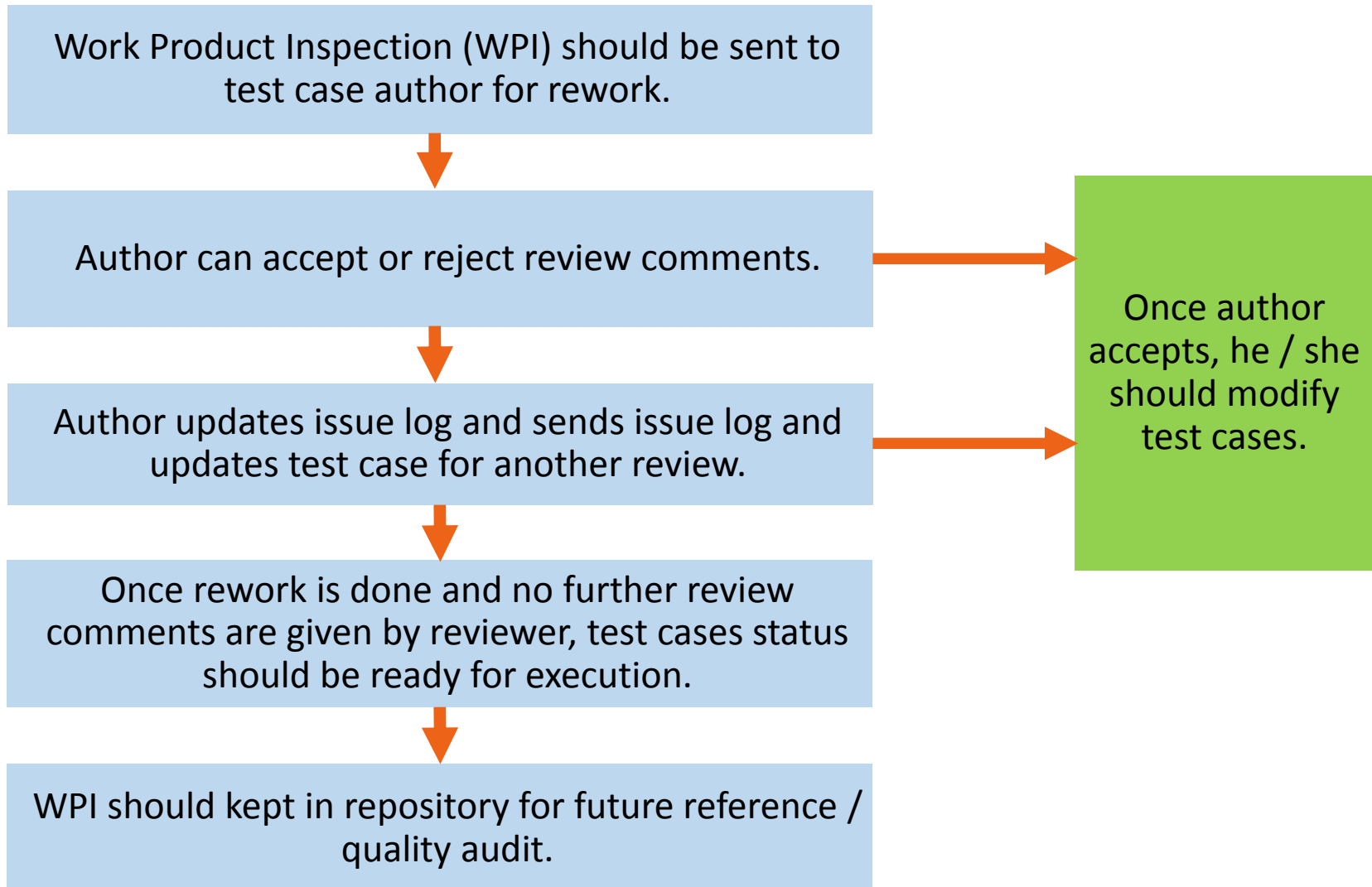


Microsoft Excel
17-2003 Workshee



Questions?

05 Idea of Rework Activities





Questions?

06 Case Study

Case Study: Day 6— RTM, Review and Rework of Test Cases



Let us get started with the some real life case studies now. Here is what you need to do:

- Work with your peers in other groups as instructed by the facilitator
- Review the test cases prepared by each other
- Record the review observations and comments directly in RQM or in OPAL template first and then enter into RQM to maintain records
- Refer the four use cases on Page 8 and 9
- Share your key takeaways with the class on the steps you took to prepare the test cases (30 mins)



Microsoft Word
17 - 2003 Document



Questions?